

Homework 2 – Calculus & Analytical Geometry

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SECTION A: Analytical Work

Q1.

$$(a) \lim_{x \rightarrow 2} (3x^2 - 5x + 4)$$

Since it's a polynomial, it's continuous on its domain. Direct substitution gives a determinate value

$$3(2)^2 - 5(2) + 4 = 6$$

$$(b) \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

Direct substitution gives an indeterminate value $\frac{0}{0}$, so

$$\frac{x^2 - 9}{x - 3} = \frac{(x+3)(x-3)}{x-3} \Rightarrow x+3$$

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} (x+3) \Rightarrow 6$$

$$(c) \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$$

Direct substitution gives an indeterminate value $\frac{0}{0}$, so

$$\frac{x^3 - 8}{x^2 - 4} \Rightarrow \frac{(x-2)(x^2 + 2x + 4)}{(x-2)(x+2)} \Rightarrow \frac{x^2 + 2x + 4}{x+2}$$

$$\lim_{x \rightarrow 2} \left(\frac{x^3 - 8}{x^2 - 4} \right) = \frac{\lim_{x \rightarrow 2} (x^2 + 2x + 4)}{\lim_{x \rightarrow 2} (x+2)} \Rightarrow 3$$

$$(d) \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$$

Indeterminate value by direct substitution so we'll rationalize first.

$$\frac{(\sqrt{x} - 2)(\sqrt{x} + 2)}{(x - 4)(\sqrt{x} + 2)} = \frac{(\cancel{x} - 2)(\cancel{x} + 2) \cancel{x} / 4}{(x - 4)(\sqrt{x} + 2)}$$

$$\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4} = \lim_{x \rightarrow 4} \frac{1}{\sqrt{x} + 2} \Rightarrow \frac{1}{2 + 2} \Rightarrow \frac{1}{4}$$

$$(e) \lim_{x \rightarrow 0} \frac{\sqrt{1 + 2x} - 1}{x}$$

Indeterminate value so we'll rationalize as:

$$= \frac{(\sqrt{1 + 2x} - 1)(\sqrt{1 + 2x} + 1)}{x(\sqrt{1 + 2x} + 1)}$$

$$= \frac{x + 2x - 1}{x + 2\sqrt{1 + 2x}} \Rightarrow \frac{2}{\sqrt{1 + 2x} + 1}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 + 2x} - 1}{x} = \lim_{x \rightarrow 0} \frac{2}{\lim_{x \rightarrow 0} \sqrt{1 + 2x} + 1} \Rightarrow \frac{2}{2} = 1$$

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$$(f) \lim_{x \rightarrow 0} \frac{\sin 5x}{x}$$

Indeterminate by direct substitution, but

$$\frac{5}{5} \cdot \lim_{x \rightarrow 0} \frac{\sin 5x}{x} \Rightarrow 5 \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \Rightarrow 5(1) \Rightarrow 5$$

$$(g) \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$$

Indeterminate by direct substitution. So:

$$\therefore 1 - \cos 2x = 2 \sin^2 x$$

$$= 2 \frac{\sin^2 x}{x^2} \Rightarrow 2 \left(\frac{\sin x}{x} \right)^2 \Rightarrow 2(1)^2 \Rightarrow 2.$$

Q 2:

$$(a) \lim_{x \rightarrow 0} f(x) = \frac{|x|}{x}$$

$$\lim_{x \rightarrow 0^-} f(x) = \frac{1-x}{-x} \Rightarrow -1$$

$$\lim_{x \rightarrow 0^+} f(x) = \frac{|x|}{x} \Rightarrow 1$$

L.H.L $\neq$ R.H.L so two sided limit does not exist.

$$(b) \quad g(x) = \frac{x+1}{x-2}$$

$$\lim_{x \rightarrow 2^-} (g(x)) = \frac{3}{2-2} \Rightarrow -\infty$$

$$\lim_{x \rightarrow 2^+} (g(x)) = \frac{3}{2-2} \Rightarrow +\infty$$

$$L.H.L \neq R.H.L$$

Graph has a vertical asymptote near $x=2$ where it goes to $-\infty$ from left and $+\infty$ from right.

$$(c) \quad \lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$$

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2 \quad \text{where } x^2 \geq 0$$

As $\lim_{x \rightarrow 0}$, by squeeze theorem, $x^2 \sin\left(\frac{1}{x}\right) = 0$

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Q 3.

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$$f(x) = \begin{cases} x^2 + k, & x < 2 \\ 3x - 1, & x \geq 2 \end{cases}$$

$$(a) \lim_{x \rightarrow 2} f(x)$$

$$\lim_{x \rightarrow 2^-} f(x) = (2)^2 + k \Rightarrow 4 + k$$

$$\lim_{x \rightarrow 2^+} f(x) = 3(2) - 1 \Rightarrow 6 - 1 \Rightarrow 5$$

$$k = 5 - 4 \Rightarrow 1$$

(b) k for $f(x)$ to be continuous at 2

$$\text{for } k = 1$$

$$\lim_{x \rightarrow 2^-} f(x) = 4 + 1 \Rightarrow 5$$

$$\lim_{x \rightarrow 2^+} f(x) = 6 - 1 \Rightarrow 5$$

L.H.L = R.H.L so $f(x)$ is continuous at $x=2$ & $k=1$.

(c) Verification:-

Since L.H.L & R.H.L are equal, there'll be no break on graph so it is continuous. The point (5) will be a filled circle instead of a hole.

4-

$$(a) p(x) = \frac{x^2 - 4}{x - 2}$$

$$p(x) = \frac{x^2 - 4}{x - 2} = x + 2 \quad \therefore x \neq 2$$

$\lim_{x \rightarrow 2} p(x) = 4$ exists but $p(2)$ is undefined.

• Hence it is a "removable discontinuity" at $x = 2$.

$$(b) q(x) = \begin{cases} x^2 - 4 & x < 1 \\ x + 3 & x \geq 1 \end{cases}$$

$$\lim_{x \rightarrow 1^-} q(x) = 1 - 4 = -3$$

$$\lim_{x \rightarrow 1^+} q(x) = 1 + 3 = 4$$

• $-3 \neq 4$ hence it is a "jump discontinuity" at $x = 1$.

$$(c) r(x) = \frac{1}{(x+2)^2}$$

$r(x)$ is undefined at $x = -2$

$$\lim_{x \rightarrow -2^-} r(x) = +\infty$$

$$\lim_{x \rightarrow -2^+} r(x) = +\infty$$

$$\therefore L.H.L = R.H.L$$

• Infinite discontinuity at $x = -2$.

4.

$$(d) \quad s(x) = \frac{x^2 - x - 6}{x^2 - 4x + 3}$$

$$\frac{x^2 - x - 6}{x^2 - 4x + 3} = \frac{(x-3)(x+2)}{(x-1)(x-3)} \Rightarrow \frac{x+2}{x-1}$$

• Hence undefined at (1, 3).

• At $x=3$, $\lim_{x \rightarrow 3} s(x) = 5/2$ but undefined at $s(3)$, "removable".

• At $x=1$, $\lim_{x \rightarrow 1^+} s(x)$ & $\lim_{x \rightarrow 1^-} s(x) = \pm \infty$, so infinite discontinuity.

Q 5.

$$R(t) = \frac{t^2 - 25}{t - 5}, \quad t \neq 5$$

(a) $R(t)$, $t \neq 5$

$$R(t) = \frac{(t+5)(t-5)}{t-5} \Rightarrow t+5$$

(b) $\lim_{x \rightarrow 5} R(t)$

$$\begin{aligned} \lim_{x \rightarrow 5} R(t) &= \lim_{x \rightarrow 5} (t+5) \\ &= 10 \end{aligned}$$

(c) Continuity:

• $\lim_{x \rightarrow 5} R(t)$ exists but $R(5)$ is undefined so the

first continuity condition fails.

(d) $R(s)$

- Defining $R(s) = 10$ makes the function continuous everywhere.

(e) Graphing Program

Since only $t=5$ is the point where the function was discontinuous, a graphing program might fail to plot a hole at this point or may pass a straight line through the point as if it was a continuous function.